

HAAD Practitioner Workpapers

Standalone forms derived from Version 0.4

Version 0.4 · Operational companion

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Classification key

Evidentiary classifications

- **Known—established:** reliable evidence establishes the proposition and materially plausible alternatives have been addressed.
- **Known—contradicted:** reliable evidence establishes that the proposed transition did not occur as alleged.
- **Assumed:** used for a limited analytical purpose but not established; the assumption and effect are explicit.
- **Undetermined—pending:** potentially resolvable evidence remains to be obtained or tested.
- **Undetermined—exhausted, absent:** the necessary evidence was not created, retained, or is established not to exist.
- **Undetermined—exhausted, inaccessible:** evidence may exist, but reasonable acquisition avenues within authority and scope are exhausted.
- **Undetermined—exhausted, existence unknown:** reasonable inquiries cannot establish whether the evidence exists.

Scope dispositions

- **Not in issue:** the dimension belongs to the model, but no material fact, context, or competing explanation places it in dispute.
- **Not applicable:** the dimension does not logically apply to the scoped attribution claim.

These are not evidentiary conclusions and do not mean known—established.

Canonical attribution dimensions

Use these identifiers and names throughout the workpapers:

ID	Canonical dimension	Discontinuity classification
AD-01	External action	Action (7.1)
AD-02	Technical actor and execution	Execution (7.2)
AD-03	Credential or account	Identity (7.3)
AD-04	Human identity	Identity (7.3)
AD-05	Natural-person control	Control (7.4)
AD-06	Authorship of the material interaction	Interaction-authorship (7.5)
AD-07	Instruction origin	Instruction-origin (7.6)
AD-08	Knowing authorization	Authorization (7.7)
AD-09	Voluntariness	Voluntariness (7.8)
AD-10	Scope	Scope (7.9)
AD-11	Knowledge	Knowledge (7.10)

ID	Canonical dimension	Discontinuity classification
AD-12	Intent	Intent (7.11)
AD-13	Delegation	Delegation (7.12)
AD-14	Causal contribution	Causal-contribution (7.17)

Individual and organizational investigative attribution are synthesized conclusions. Responsibility referral is a downstream output.

Discontinuity classification

Record:

1. the affected attribution dimension;
2. the failure condition or conditions;
3. the graph location; and
4. whether the discontinuity is primary, co-primary, contributing, causal, or downstream.

Matter control sheet

Field	Entry
Matter identifier	
Methodology version	Version 0.4
Practitioner	
Independent reviewer	
Date opened	
Application level	Triage / Standard / Enhanced
Mode	Forensic / Assurance
Evidence-preservation location	
Workpaper access classification	

WP-01 Attribution Question and Scope Record

Field	Entry
Consequential action	
Proposed human or organizational attribution	
Attribution dimensions in issue	

Field	Entry
Period and material time	
Systems and trust domains	
Included questions	
Excluded questions	
Mode: forensic or assurance	
Application level and justification	
Dimensions not in issue	
Dimensions not applicable	
Individual path, organizational path, or both	

WP-02 Attribution Graph and Transition Register

Transition ID	From	To	Proposition	Material to which conclusion	Expected evidence	Trust-boundary crossing	Topology or contribution role
T-01							

Attach the graph or identify its preservation location:

Field	Entry
Graph location	
Topology	Hierarchical / peer-to-peer / mesh / hybrid / undetermined
Material forks, merges, or feedback loops	
Stored-state crossings	
Trust transformations	

WP-03 Proposition-Specific Evidence and Independence Record

Complete one row for every proposition–stream relationship. The same record may be independent for one proposition and dependent or irrelevant for another.

Proposition	Stream ID	Evidence and independent face fact	Integrity and completeness	Capture mechanism	Dependencies or inherited fields	Control plane or custodian	Common failure path	Evidentiary contribution and limit
	S-01							

Single-stream known-established exception

Complete only where a ceiling-setting proposition is proposed as known-established without two proposition-relevant streams.

Proposition	Reason second stream is unavailable or unnecessary	Direct observation or inherited assertion	Alteration capability	Expected confirming or contradictory records	Alternatives tested	Residual failure risk	Why classification is not lowered	Second-practitioner concurrence, if required

WP-04 Observation, Inference, and Warrant Log

Entry	Observation	Proposed inference	Warrant required	Support for warrant	Assumption or limit
OI-01					

For testimonial evidence, decompose admissions, denials, recollection failures, and explanations into separate propositions.

Statement ID	Statement made	Proposition asserted	Speaker's interest	Contemporaneity and specificity	Consistency	Independent corroboration	Evidentiary treatment
TS-01							

WP-05 Competing-Origin and Intervention Matrix

An alternative must pass both the plausibility and materiality gates.

Transition ID	Hypothesis	Plausibility anchor	Material effect if accepted	Originating source	Immediate influencing source	Content or trust transformation	Supporting evidence	Contrary or missing evidence	Status
T-01									

WP-06 Transition Adjudication Matrix

Transition ID	Proposition	Evidence relied upon	Proposition-specific convergence	Zemi classification or scope disposition	Exhaustion basis, if applicable	Reason	Effect on attribution
T-01							

Exhaustion record

Complete before assigning an exhausted status.

Proposition	Evidence sought	Systems and custodians checked	Material-time and scope coverage	Acquisition steps and authority	Provider or witness responses	Why further inquiry is unavailable or disproportionate	Exhausted state selected

WP-07 Discontinuity Schedule

HAAD ID	Canonical dimension: AD ID and name	Failure condition	Role: primary, co-primary, contributing, causal, or downstream	Graph location	Resolution status	Remaining alternatives	Conclusion ceiling	Required future evidence
H-01								

Primary discontinuity:

Field	Entry
Scoped attribution proposition	
Earliest independently ceiling-setting transition on that causal path	
Why downstream evidence cannot cure it	
Co-primary transition, if any	

WP-08 Bounded Attribution Conclusion

Conclusion layer	Finding	Classification or scope disposition	Attribution boundary and basis
External action			
Technical actor and execution			

Conclusion layer	Finding	Classification or scope disposition	Attribution boundary and basis
Credential or account			
Human identity			
Natural-person control			
Authorship of the material interaction			
Instruction origin			
Knowing authorization			
Voluntariness			
Scope			
Knowledge			
Intent			
Delegation			
Causal contribution			
Individual investigative attribution			
Organizational investigative attribution			
Responsibility referral			

Required bounded conclusion:

Field	Entry
Nearest established actor or state	
Transition at which attribution stops	
Weakest material transition	
Conclusion that would exceed the evidence	
No demonstrated discontinuity within scope, if applicable	
Express limitations	

WP-09 Assurance and Readiness Return

Discontinuity	Evidence that was absent or unreliable	Control or instrumentation requirement	Owner	Priority	Reassessment trigger
H-01					

WP-10 Organizational Attribution Path

Complete for Enhanced application and whenever organizational attribution is in scope.

Transition ID	Organizational actor or function	Decision, authority, notice, control, or enforcement proposition	Evidence	Classification	Effect on organizational attribution	Readiness implication
OT-01						

Quality-control sign-off

Check	Practitioner	Reviewer
External action established independently of the subject agent's assertion		
Technical identity not treated as natural-person identity		
Device or session control not treated as authorship of every material interaction		
Technical permission not treated as substantive authorization		
Authorization distinguished from scope		
Downstream evidence not used to cure an upstream attribution break		
Every stream counted for a named proposition		
Single-stream exception fully documented and concurred with where required		
Testimonial assertions decomposed and corroboration assessed		
Competing alternatives passed both gates		
Affected dimensions distinguished from failure conditions		
Pending and exhausted states supported by the required inquiry record		
DRT not treated as a waiver or transfer of organizational control responsibility		
Individual and organizational paths adjudicated separately		
Report states the exact attribution boundary or no demonstrated discontinuity within scope		

Sign-off field	Entry
Practitioner name and date	
Reviewer name and date	
Unresolved disagreement	
Required escalation or revision	